Sustainable Water Management Systems

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Sustainable Water Management Systems (A13241)

Short Title:	Sustainable Water Mgmt Systems		
Faculty	Science and Computing		
Department	Science		
Cluster	Environmental Science		
Author	JGERAGHTY		
Credits	5	**Level:**	Advanced

Description of Module / Aims

The demands of population increases, associated food requirements and the influence of climate change will greatly impact on water availability and associated quality. These issues are going to place additional land and water management demands on professionals in agriculture, forestry and horticulture. This module will provide participating students with a solid foundation in the guiding principles of sustainable water management and use. Students will be able to assess water needs and requirements for animals and crops, including trees, under different situations, and design appropriate irrigation or drainage systems. Innovative flood mitigation and adaptation measures on land, in river catchments and coastal areas will also be reviewed and analysed.

Programmes

`SCIE-0072`	BSc (Hons) in Agricultural Science (`WD_SAGRI_B`)	4 / 8 / E
`SCIE-0072`	BSc (Hons) in Land Management (in Agriculture) (`WD_SBSAG_B`)	1 / 2 / E
`SCIE-0072`	BSc (Hons) in Land Management (in Forestry) (`WD_SBSFO_B`)	1 / 2 / E
`SCIE-0072`	BSc (Hons) in Land Management (in Horticulture) (`WD_SBSHO_B`)	1 / 2 / E

Indicative Content

- Global issues surrounding water resources, climate change, water availability, quality management
- Soil and water: evaporation, hydraulic capacity, infiltration, moisture content, percolation, rainfall effects, soil texture
- Irrigation: methods, system and structure design, crop requirements, salinity and sodicity
- Subsurface drainage: principles, design, equipment, practice, performance and evaluation
- River management: water flow rates, spate flooding, managing riparian zones and river catchments
- Wetlands: water attenuation, regenerating peat lands, functional landscape approaches
- Coastal issues: rising sea levels, river delta adaptation measures, coastal defences

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Determine priority issues for water resources and associated threats.
2. Estimate and establish drainage and irrigation requirements, rainfall amounts, storage capacity, water flow rates.
3. Design appropriate and sustainable irrigation systems and subsurface drainage systems for land water management.
4. Determine the key factors surrounding river catchment management.
5. Integrate water as an essential component in land management practice.
6. Propose a sustainable water management system for agriculture, horticulture or forestry production.
7. Produce a report on the design and development of a river catchment management plan.

Learning and Teaching Methods

- Lectures.
- Field visits.
- E-learning.
- Briefings from sector professionals.
- Webcast presentations.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,3,4,5
Continuous Assessment	40%	
Assignment	20%	6
Group Project	20%	7

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Unable to analyse key topics in the course.

40%-49%: Demonstrate a good knowledge of the key learning outcomes of the course with basic evaluation of aspects.

50%-59%: Demonstrate a good level of knowledge with aptitude to integrate and critically evaluate the various components of the course.

60%-69%: Demonstrate more detailed knowledge of the key topics including critical evaluation the roles and impacts of water management methods and practices.

70%-100%: Demonstrate a comprehensive knowledge and understanding of all learning outcomes, including the ability to integrate and critically analyse the various components of the course.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
E-Learning	12	
Field Trips	12	
Independent Learning	75	

Essential Material

- Nijland, H.J., F.W. Croon and H.P. Ritzema. _Subsurface Drainage Practices - Guidelines for the implementation, operation and maintenance of subsurface pipe drainage systems_. Wageningen, The Netherlands: International Institute for Land Reclamation and Improvement - Publication 60, 2005.
- Ritzema, H.P. _Drainage Principles and Applications_. 3rd ed. Wageningen, The Netherlands: International Institute for Land Reclamation and Improvement - Publication 16, 2006.